(PROBABLY ALMOST)

EVERYTHING ON THE SYLLABUS

XBI11 IAL EDEXCEL BIOLOGY

UNIT 1 & 2

Stem Cells & Epigenetics

Stem Cells

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Stem Cells

- Define Stem Cells

  Stem cells are **undifferentiated cells** that **continually divide** (via mitosis) to **become specialised.**

  Undifferentiated - They have no specialised structures, features or functions. They have yet to develop into a particular cell variant.

  The **continually divide** via mitosis - Allows stem cells to 1. self renew (produce identical copies of itself) and 2. form specialised cells through differentiation. This ensures that the body **maintains a stem cell pool and can replace lost/damaged specialised cells.**

  Become Specialised - Can undergo differentiation and express certain genes to become specific specialised cells.

- Differentiate between the 4 Types of Stem Cells

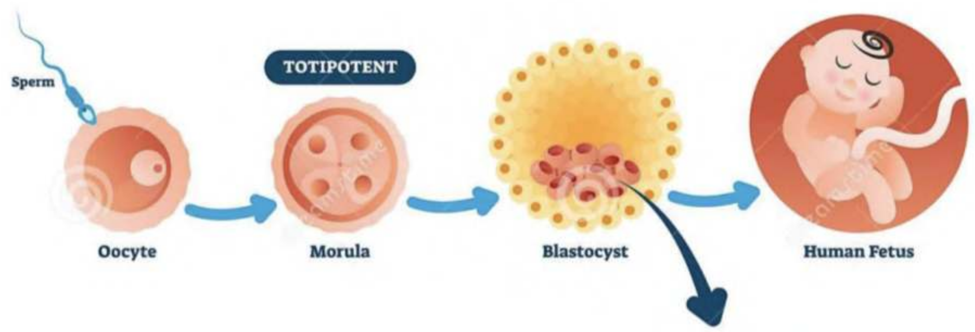
  Differences

Totipotent	Pluripotent	Multipotent	Unipotent

Totipotent	Pluripotent	Multipotent	Unipotent
Can differentiate to form any type of cells in the body (extra- and embryonic)	Can differentiate to form any type of cell except placental cells.	Can differentiate to form only closely related/family of cells. I.E. Blood stem cell → RBC, WBC, Platelets, etc.	Can divide into 1 type of cell. Helps in cell renewal. I.E. Skin stem cells only produce skin cells
Found in the early stages of the mammalian embryo, 4-5 days of development - in the morula (16 totipotent cells)	Found in the Blastocyst - after 4-5 days of fertilisation	Found throughout the body, in the bone marrow, adipose tissue, cardia cells, etc.,	Found in the bone marrow
Untitled			

- Totipotent - Cells than can divide & differentiate to embryonic & extra-embryonic cells.
- Pluripotent - Can divide & differentiate into more than 200 types other than placenta cells.
- Multi-potent Cells: Can differentiate to closely related/family of cells. Such as circulatory system cells.
- Unipotent - Can divide into 1 type of cell. Involved in cell renewal.

▼ Describe Stem Cell Formation using the human development model



When the sperm fertilises the oocyte, it forms the zygote which is a single totipotent cell. As the zygote develops it divides, which means that the stem cells all divide and increase in number. Approximately 4-5 days after the initial fertilisation, the zygote develops into the morula. The morula contains 16 cells that are all totipotent and can differentiate to form any kinds of cells including placental cells. This is the last stage at which totipotent cells are found.

After 4-5 days of fertilisation, the morula develops into the blastocyst which contains a mass of 200-300 cells including an inner cell mass that later develops into the embryo. Cells present in the blastocyst are all pluripotent.

Multipotent cells form when the pluripotent cells in the cell mass differentiate and develop into specialised, multipotent stem cells.

Unipotent stem cells form when multipotent stem cells differentiate.

- Differentiate between embryonic, adult and umbilical cord stem cells

Differences

Embryonic Stem Cells	Adult Stem Cells	Umbilical Cord Stem Cells
Found in the earliest stages of mammalian embryo development. Totipotent cells in the morula (4-5 days), pluripotent cells in the blastocyst (5-6 days)	Found in differentiated adult, body tissues and organs (e.g. bone marrow)	Found in the blood that drains from the placenta and umbilical cord after birth.
Can differentiate into almost any human cell	Can only differentiate into a few types of cells	Can differentiate into almost any human cell
Totipotent & Pluripotent	Multipotent (and Unipotent)	Pluripotent

- Outline the uses of stem cell

Is the use of stem cells to treat or prevent diseases. The practise involves taking stem cells from the body (the bone marrow, adipose tissues, early embryos) and injecting it into damaged areas to repair them. I.E. Harvesting stem cells from the bone marrow and injecting it into hearts damaged by heart attacks to repair it.

Benefits of Stem Cell Therapy

- Treating Parkinson's disease: Transplanting stem cells into the brain to produce dopamine neurones
- Treating Type 1 Diabetes: Inject stem cells to repair the pancreas to restore insulin production.
- Producing Organs for Transplants: Stem cells can differentiate to form cells and eventually functioning organs.

Pros & Cons

Pros	Cons
Although the areas of success for stem cell therapy is limited, successful treatments is expected to increase dramatically in the future.	Still hard to control the differentiation of stem cells so may cause cancer.
Poses better alternatives to other medical procedures	Chances of tissue rejection
Can save and improve the quality of life for many	Ethical concerns over using embryonic stem cells as embryos may be killed in the process.
Untitled	Risk of infection during transplantation.

▼ How do monitoring authorities regulate the use of embryonic stem cells in research

1. Monitors research to ensure that research is necessary
2. Issues licences/gives permission for stem cell research
3. Monitors sources of stem cells
4. Ensures that only early stage embryos are used as sources of stem cells
5. Prevents unethical uses of stem cell i.e. human cloning and genetic manipulation

Cell Specialisation

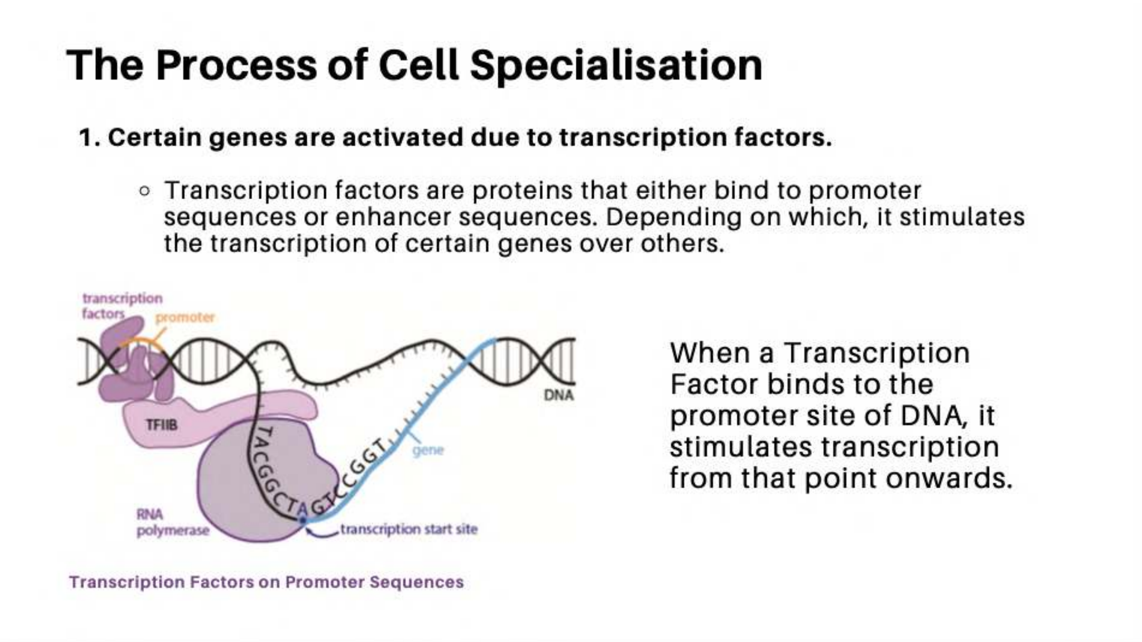
▼ Describe how Gene Expression leads to the formation of different specialised cells

- All stem cells have the complete set of DNA of an organism.

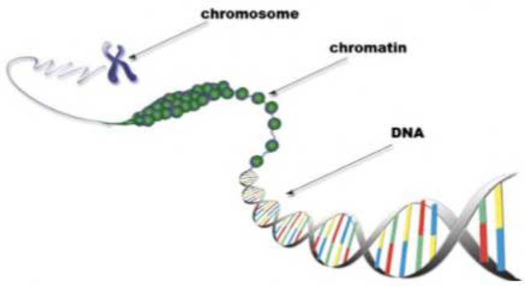
- This DNA contains all the genes necessary to make the proteins required to give a cell its specialised structure and function.
- To produce a certain specialised cell, the stem cell needs to express the specific genes required to code for the proteins necessary in that cell.
- I.E The stem cell needs to express the gene coding for Haemoglobin Protein in order to produce a RBC.

Steps:

1. Chemicals such as signal proteins trigger the activation of some genes over others.
2. Thus, some genes are active and some are inactive.
3. The active genes are transcribed to mRNA, whereas the inactive ones are not.
4. Thus, when the mRNA is translated, they produce specific proteins and enzymes which determine the structure and function of the cell.



The Process of Cell Specialisation Continued...



Transcription Factors on Enhancer Sequences

When a Transcription Factor binds to the enhancer site of DNA, it alters the structure of chromatin, making it more or less open to RNA polymerase.

Open chromatin structures allow gene expression whereas closed chromatin structures do not.

2. Genes that are not needed remain unactivated.

The Process of Cell Specialisation Continued...

3. The genes that are activated undergo **transcription to form mRNA**. The **mRNA then moves to the ribosome to be translated into polypeptides.**

4. The necessary proteins are produced and give the cell its specialised structure and function / controls the cell's processes.

    - I.E. Production of Haemoglobin consequently gives RBC the ability to transport oxygen.

5. Cell becomes Specialised. The changes are irreversible - a specialised cell can not revert back to a stem cell. At this point the cell is so specialised it loses its potency and cannot differentiate to produce cells.

▼ Explain the process of RNA Splicing

Right after transcription, mRNA (aka pre-mRNA) undergoes a process known as splicing. This is because genes are not continuous.

- To modify the mRNA, the introns (non-coding part of the gene) and some exons are removed.
- Enzyme complexes known as Spliceosomes join together the exons.
- The spliceosomes can join the same exons in a number of ways which produces different versions of mRNA from the same strand.

- This leads to the production of multiple types of proteins.

Epigenetics

▼ Explain how phenotype is the result of an interaction between genotype and the environment

Phenotypes are the characteristics expressed due to genetics and the environment. Some characteristics are only affected by genes i.e. eye colour, however, others such as skin colour, can be affected by genes: the natural colour of your skin, and the environment: sunlight, can affect it.

▼ What is epigenetics?

Epigenetics is the study of **changes of gene expression due to the environment**, without a change in the bases of DNA, but by the addition of chemical groups that affect how easy it is to transcribe genes.

▼ Explain how Epigenetic modification can alter the activation of certain genes

1. Methylation of DNA
    - Involves adding methyl groups (CH_3) to adenine or cytosine on DNA
    - Causes the chromatin/nucleosome to condense such that it is harder for RNA polymerase to bind to the region so the gene is inactive/transcribed less.
    - Is an enzyme regulated process
    - Hypermethylation (adding too many methyl groups) turns genes off, whereas hypomethylation (too few methyl groups attached) activates genes.
2. Acetylation of Histone Proteins
    - Involves adding the acetyl group to an amino acid in a histone tail
    - Promotes transcription by opening/unraveling the chromatin structure of DNA. When chromatin is more open, it allows RNA polymerase to more readingly bind to the region such that transcription can occur. Thus the gene is copied as mRNA leading to translation of its proteins,

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